

Introduction to Machine Learning

What is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computer systems to learn from data without being explicitly programmed. Instead of following rigid rules written by humans, ML systems identify patterns in historical examples and generalize those patterns to make predictions on new, unseen data. This paradigm shift has made it possible to solve problems that are difficult to codify by hand, such as image recognition, natural language understanding, and fraud detection.

The core idea is simple: given a dataset of examples, a learning algorithm adjusts an internal mathematical model so that its predictions become progressively more accurate. The quality of the model depends on three factors: the quantity and quality of the training data, the expressiveness of the model architecture, and the choice of the optimization procedure. More data, better features, and stronger models generally lead to better performance, but they also increase computational cost.

Machine learning pipelines are typically divided into a training phase and an inference phase. During training, the model is exposed to labeled examples and its parameters are updated iteratively. During inference, the trained model is deployed to production where it processes real-world requests, often with strict latency requirements.

Common Algorithms

Algorithms in machine learning fall into three broad families. Supervised learning uses labeled data: classification assigns inputs to discrete categories, while regression predicts continuous values. Popular supervised methods include linear and logistic regression, decision trees, random forests, and gradient-boosted trees.

Unsupervised learning finds structure in unlabeled data. Clustering algorithms such as k-means group similar points together, while dimensionality reduction techniques like PCA and t-SNE compress high-dimensional data into a smaller number of meaningful dimensions. These methods are widely used for exploratory analysis and anomaly detection.

Reinforcement learning trains agents through interaction: the agent takes actions in an environment, receives rewards or penalties, and learns a policy that maximizes cumulative reward over time. Reinforcement learning powers breakthroughs in robotics, game playing, and autonomous systems.

Applications

Machine learning now underpins products used by billions of people. Email providers use classifiers to filter spam, streaming platforms use recommendation systems to surface content, and banks deploy fraud-detection models that score transactions in real time. In healthcare, models assist radiologists by highlighting suspicious regions in medical scans, improving both speed and accuracy of diagnosis.

Large language models, a recent milestone, apply deep learning at massive scale to text. They are trained on trillions of tokens and can summarize documents, answer questions, translate languages, and generate code. When augmented with retrieval systems — a technique known as retrieval-augmented generation, or RAG — they can ground their answers in specific documents, dramatically reducing hallucinations.

Deploying machine learning responsibly requires attention to data privacy, model fairness, and interpretability. Teams must audit models for bias, protect sensitive training data, and document how predictions are made so that stakeholders can trust the system. These considerations are now a standard part of the ML engineering lifecycle.